Cancer Type Classification from Gene Expression Data Using Autoencoder-Based Dimensionality Reduction

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MSE 5760 Machine Learning And Its Applications In Materials Science

[University of Pennsylvania]

May 9, 2026

Abstract

High-dimensional gene expression data from RNA sequencing presents significant challenges for cancer type classification due to the curse of dimensionality. In this study, we investigate autoencoder-based dimensionality reduction methods for classifying 33 cancer types using the TCGA Pan-Cancer Atlas RNA-Seq dataset comprising 11,060 tumor samples and 20,531 genes. We implement and compare three dimensionality reduction approaches: a Stacked Denoising Autoencoder (SDAE), a Variational Autoencoder (VAE), and Principal Component Analysis (PCA) as a linear baseline. After reducing the gene expression data from 5,000 selected high-variance genes to a 256-dimensional latent representation, we train identical fully connected neural network classifiers on each set of extracted features. Our results show that PCA achieves the highest overall accuracy of 94.67%, followed by VAE at 93.17% and SDAE at 92.86%. All three methods successfully classify most cancer types with high accuracy, though certain molecularly similar cancers—such as colon and rectum adenocarcinoma—remain challenging to distinguish. An ablation study over latent dimensions (32–512) reveals that classification performance is relatively stable across different bottleneck sizes. These findings suggest that for this particular classification task, linear dimensionality reduction via PCA captures sufficient discriminative information, while the nonlinear autoencoder methods offer advantages in latent space structure that may benefit other downstream tasks such as clustering and data generation.

1 Introduction

Cancer is one of the leading causes of death worldwide, and accurate molecular classification of tumors is essential for effective diagnosis and personalized treatment planning. While traditional histopathological methods rely on morphological characteristics, molecular profiling through high-throughput RNA sequencing (RNA-Seq) enables characterization of tumors at the transcriptomic level, providing a more comprehensive view of the underlying biology.

The Cancer Genome Atlas (TCGA) project has profiled over 11,000 tumor samples across 33 cancer types, generating standardized gene expression measurements for more than 20,000 genes per sample [5]. This rich dataset presents both an opportunity and a computational challenge: the extremely high dimensionality of gene expression data (with far more features than samples) leads to the well-known curse of dimensionality, making direct classification difficult and computationally expensive.

Dimensionality reduction is therefore a critical preprocessing step. Traditional linear methods such as Principal Component Analysis (PCA) project data onto directions of maximum variance, but may fail to capture complex nonlinear relationships inherent in biological data. Deep learning-based approaches, particularly autoencoders, have emerged as powerful tools for learning compressed, low-dimensional representations that can capture biologically meaningful features.

In this project, we compare three dimensionality reduction strategies for pan-cancer classification:

1. **Stacked Denoising Autoencoder (SDAE):** An autoencoder trained to reconstruct clean inputs from corrupted versions, encouraging robust feature extraction [2].
2. **Variational Autoencoder (VAE):** A generative model that learns a probabilistic latent space, enabling both feature extraction and data generation [3].
3. **PCA:** A classical linear baseline for comparison.

Our approach follows prior work by Chen et al. [1], who demonstrated 98% cancer detection accuracy using deep autoencoders on TCGA data, and Way and Greene [3], who applied a VAE (named "Tybalt") to extract biologically meaningful latent features from TCGA gene expression profiles. We extend these studies by systematically comparing SDAE, VAE, and PCA under identical experimental conditions and analyzing the effect of latent dimensionality on classification performance.

2 Dataset

We used the TCGA Pan-Cancer Atlas RNA-Seq gene expression dataset, downloaded from the UCSC Xena data portal [5]. The dataset was generated by the TCGA Pan-Cancer Atlas project and has been normalized for batch effects across cancer cohorts. The key characteristics are:

- **Samples:** 11,060 tumor tissue samples after alignment and filtering
- **Genes:** 20,531 gene expression values per sample
- **Cancer types:** 33 distinct categories
- **Expression values:** Batch-corrected, $\log_2$-transformed normalized RNA-Seq counts measured by the Illumina HiSeq platform

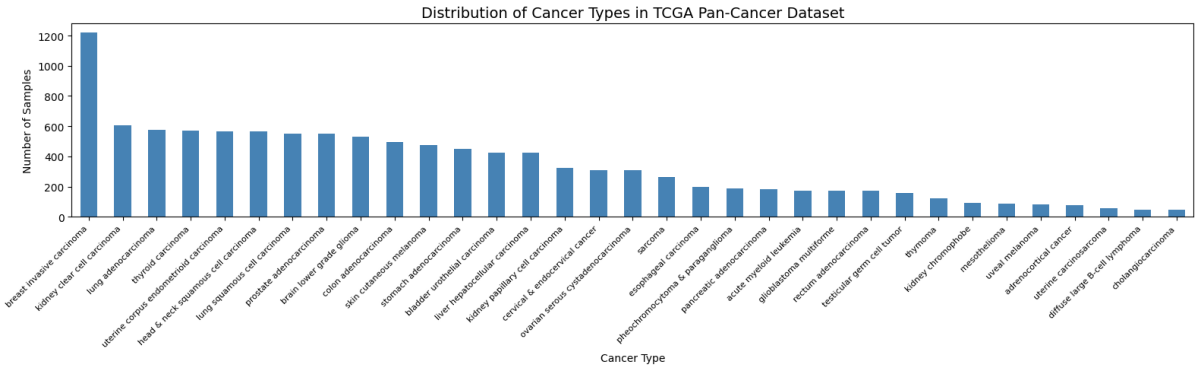

Figure 1: Distribution of cancer types in the TCGA Pan-Cancer Atlas dataset. The dataset exhibits significant class imbalance, with breast invasive carcinoma being the most represented (1,218 samples) and cholangiocarcinoma the least (45 samples).

As shown in Figure 1, the dataset exhibits considerable class imbalance. Breast invasive carcinoma accounts for the largest proportion with 1,218 samples, while rare cancers such as cholangiocarcinoma (45 samples) and diffuse large B-cell lymphoma (48 samples) have substantially fewer representatives. This imbalance is an important consideration when interpreting per-class classification accuracy.

3 Model Architecture

Our approach consists of two stages: (1) unsupervised dimensionality reduction to extract latent features, and (2) supervised classification using the learned representations.

3.1 Data Preprocessing

The raw gene expression matrix contained 3,971,737 missing values, which were imputed with zeros. We then selected the top 5,000 most variable genes by variance across samples, following the approach of Way and Greene [3]. This filtering step removes uninformative genes with low variability while retaining the most discriminative features. The selected gene expression values were standardized using z-score normalization (zero mean, unit variance). Labels were encoded using integer encoding across all 33 cancer types, and the dataset was split into 80% training (8,848 samples) and 20% test (2,212 samples) sets using stratified sampling to preserve class proportions.

3.2 Stacked Denoising Autoencoder (SDAE)

The SDAE is an autoencoder that adds Gaussian noise ($\sigma = 0.3$) to the input during training and learns to reconstruct the original clean input. This denoising objective forces the network to learn robust, noise-invariant features rather than memorizing the data. The architecture is:

Table 1: SDAE architecture (22,276,768 total parameters).

Layer	**Dimensions**	**Activation**
Input (+ Gaussian noise)	5,000	—
Encoder Layer 1	2,000	BatchNorm + ReLU + Dropout(0.2)
Encoder Layer 2	500	BatchNorm + ReLU + Dropout(0.2)
Latent (Bottleneck)	256	BatchNorm + ReLU
Decoder Layer 1	500	BatchNorm + ReLU
Decoder Layer 2	2,000	BatchNorm + ReLU
Output	5,000	Linear

The SDAE was trained for 100 epochs using the Adam optimizer (learning rate 10^{-3}, weight decay 10^{-5}) with mean squared error (MSE) reconstruction loss. A learning rate scheduler with patience of 5 epochs and a reduction factor of 0.5 was applied.

3.3 Variational Autoencoder (VAE)

The VAE learns a probabilistic latent space by encoding inputs into a mean vector $\boldsymbol{\mu}$ and log-variance vector $\log \boldsymbol{\sigma}^2$, from which latent codes are sampled via the reparameterization trick: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \exp(0.5 \cdot \log \boldsymbol{\sigma}^2)$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The VAE encoder shares the same layer dimensions as the SDAE but replaces the final encoder layer with two parallel linear layers for $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$ (22,404,512 total parameters).

The VAE loss function combines reconstruction loss and KL divergence:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \cdot D_{KL}\big(q(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big) \tag{1}$$

where $\beta = 0.5$ was used to balance the two loss terms. The VAE was trained for 100 epochs with the same optimizer settings as the SDAE. For downstream classification, we use the mean vector $\boldsymbol{\mu}$ as the latent representation.

3.4 PCA Baseline

As a linear baseline, we applied PCA to reduce the 5,000-dimensional gene expression data to 256 principal components. The top 256 components captured 82.06% of the total variance, indicating that a substantial portion of variability in gene expression can be explained by linear combinations of genes.

3.5 Classifier

An identical fully connected classifier was trained on each set of 256-dimensional latent features:

Table 2: Classifier architecture.

Layer	Dimensions	Activation
Input (Latent)	256	—
Hidden Layer 1	128	BatchNorm + ReLU + Dropout(0.3)
Hidden Layer 2	64	BatchNorm + ReLU + Dropout(0.3)
Output	33	Softmax

Each classifier was trained for 100 epochs using cross-entropy loss with the Adam optimizer (learning rate 10^{-3}, weight decay 10^{-4}).

4 Results

4.1 Autoencoder Training

Both autoencoders converged smoothly over 100 training epochs. The SDAE reconstruction loss (MSE) decreased from 0.52 to 0.21 (Figure 2), indicating that the model learned meaningful compressed representations despite the input noise corruption. The VAE total loss decreased from approximately 2,950 to 947, with the reconstruction component dropping from 2,800 to 886 and the KL divergence stabilizing around 122 (Figure 3). The KL divergence curve shows an initial increase (epochs 1–15) followed by a gradual decrease, reflecting the typical VAE training dynamic where the model first expands its latent space usage before regularizing it.

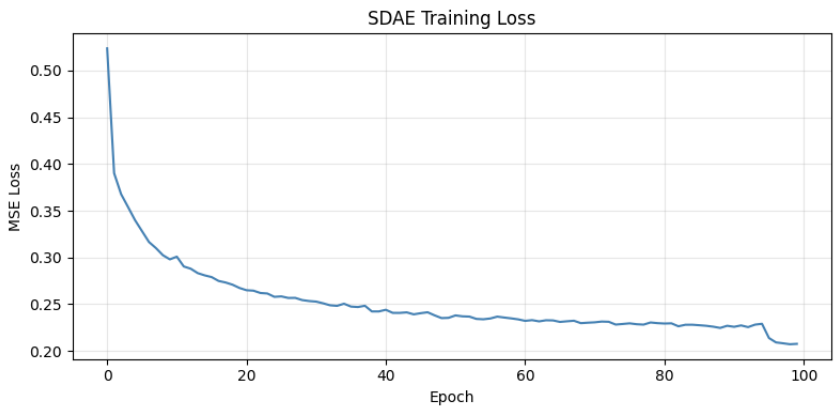

Figure 2: SDAE training loss (MSE) over 100 epochs.

Figure 3: VAE training losses: total loss (left), reconstruction loss (center), and KL divergence (right).

4.2 Latent Space Visualization

Figure 4 shows UMAP projections of the 256-dimensional latent representations from each method, using 3,000 randomly sampled training points. All three methods produce clearly separated clusters corresponding to different cancer types, confirming that the learned representations capture biologically meaningful structure.

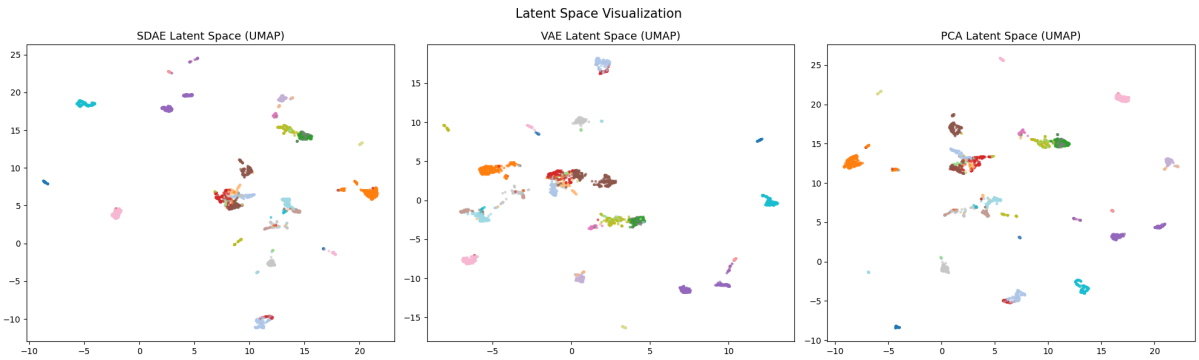

Figure 4: UMAP visualization of latent spaces learned by SDAE, VAE, and PCA. Each color represents a different cancer type. All methods produce well-separated clusters.

The SDAE latent space shows the tightest, most compact clusters, while the VAE produces more dispersed but still well-separated groupings. The PCA latent space shows clear separation for most cancer types, with some overlap in the central region where molecularly similar cancers are projected close together.

4.3 Classification Performance

Table 3 summarizes the classification performance of all three methods on the held-out test set (2,212 samples).

Table 3: Classification performance comparison on 33 cancer types.

Model	**Accuracy**	**F1 (Macro)**	**F1 (Weighted)**
SDAE + Classifier	0.9286	0.8990	0.9266
VAE + Classifier	0.9317	0.9010	0.9300
PCA + Classifier	**0.9467**	**0.9203**	**0.9457**

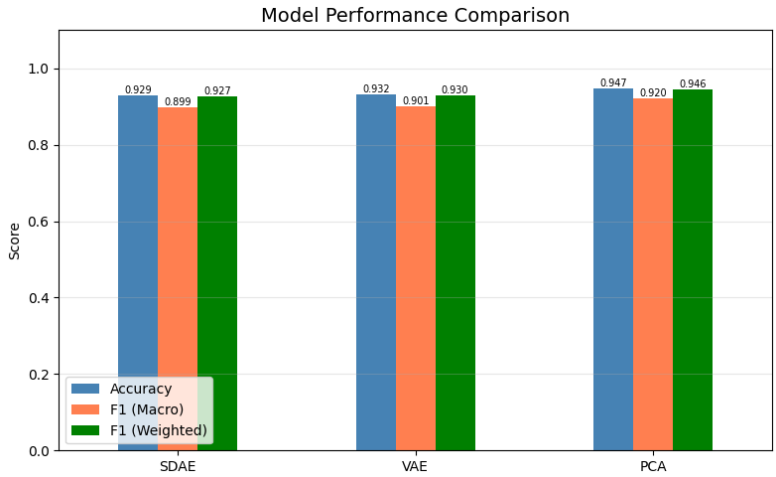

Figure 5: Model performance comparison across three metrics.

PCA achieved the best performance across all three metrics, with an accuracy of 94.67% and macro F1-score of 0.9203. The VAE slightly outperformed the SDAE (93.17% vs. 92.86% accuracy). The gap between macro and weighted F1-scores (approximately 0.02–0.03) reflects the impact of class imbalance, where rare cancer types with lower accuracy are given equal weight in the macro average.

Figure 6 shows the training dynamics of the three classifiers. The PCA classifier converges fastest and reaches the highest test accuracy, likely because PCA features, being decorrelated principal components, provide a more optimization-friendly input space. The SDAE and VAE classifiers converge more slowly but reach stable performance by epoch 40–60.

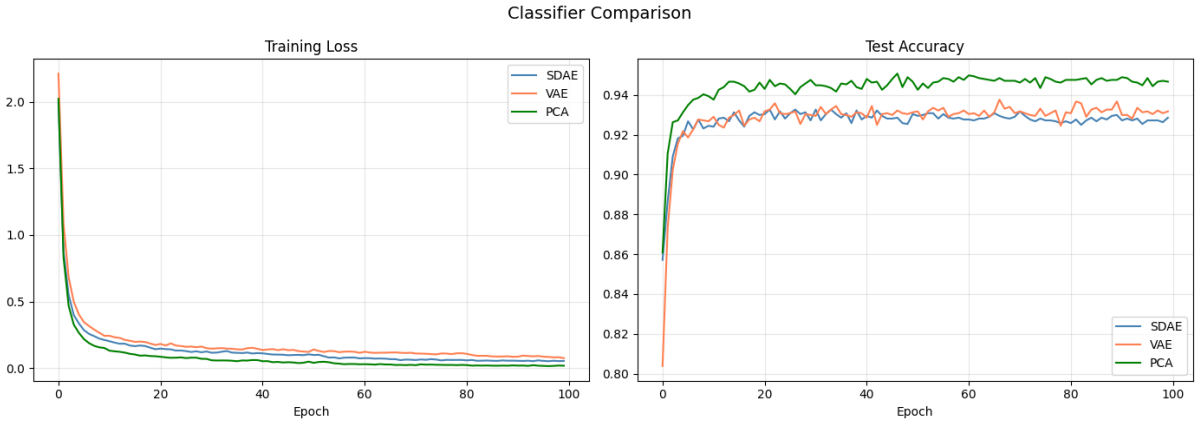

Figure 6: Classifier training loss (left) and test accuracy (right) over 100 epochs.

4.4 Confusion Matrix Analysis

Figure 7 shows the normalized confusion matrices for all three models. The strong diagonal pattern across all models indicates high classification accuracy for most cancer types.

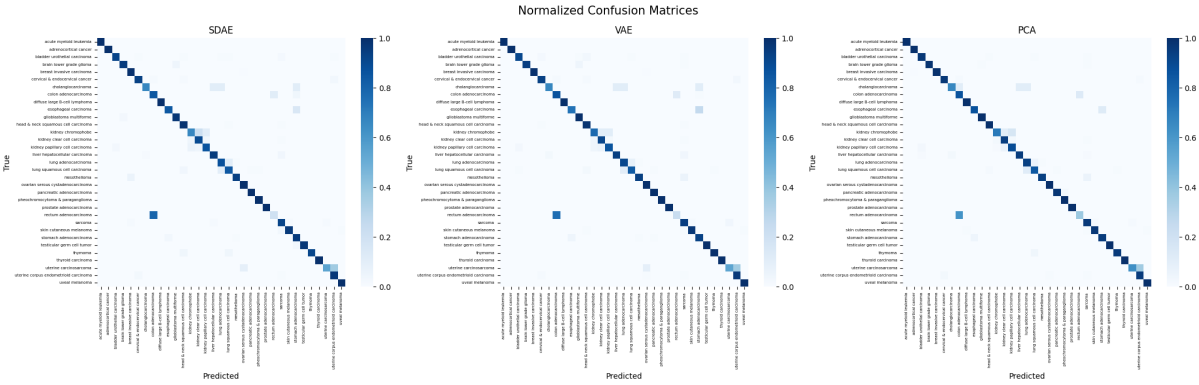

Figure 7: Normalized confusion matrices for SDAE, VAE, and PCA classifiers.

4.5 Per-Class Accuracy

Figure 8 presents the per-class accuracy for each method across all 33 cancer types. Several cancer types achieve near-perfect classification (accuracy ≥ 0.98) across all methods, including acute myeloid leukemia, breast invasive carcinoma, prostate adenocarcinoma, testicular germ cell tumor, and thyroid carcinoma. These cancers have highly distinctive gene expression profiles that are easily separable in the reduced feature space.

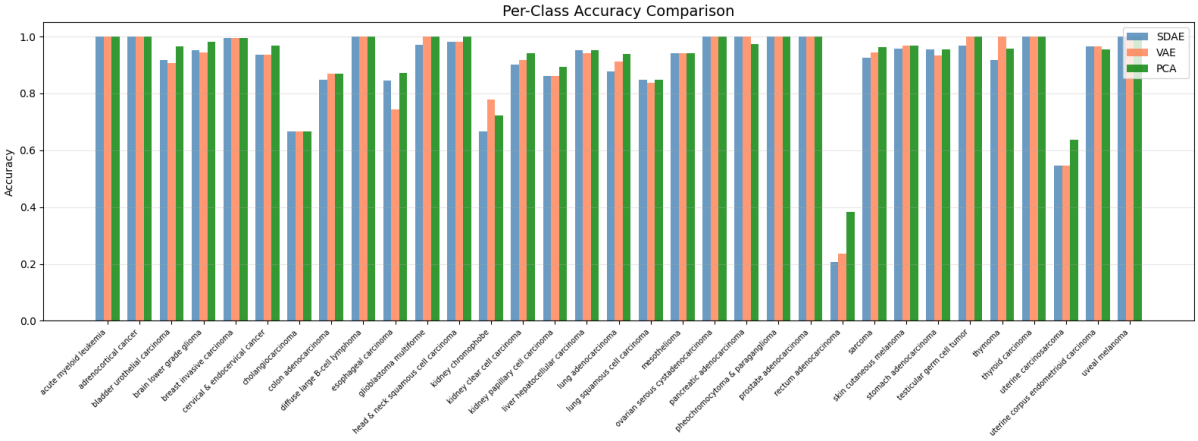

Figure 8: Per-class accuracy comparison across the three dimensionality reduction methods.

The most challenging cancer types for classification include:

- **Rectum adenocarcinoma:** Only 38–50% accuracy across methods, consistently confused with colon adenocarcinoma. This is biologically expected, as these two cancers originate from the same tissue type (colorectal epithelium) and share highly similar molecular profiles.
- **Cholangiocarcinoma:** 67–75% accuracy, likely due to its small sample size (only 45 total, 9 in test set) and molecular overlap with liver hepatocellular carcinoma.
- **Uterine carcinosarcoma:** 55–88% accuracy, with confusion against uterine corpus endometrioid carcinoma due to shared tissue of origin.
- **Kidney chromophobe:** 72–76% accuracy, with some confusion against other kidney cancer subtypes.

The best-performing model (PCA) detailed classification report shows precision, recall, and F1-scores above 0.90 for 25 out of 33 cancer types, demonstrating strong overall performance despite the challenges posed by molecularly similar cancers and class imbalance.

4.6 Ablation Study: Effect of Latent Dimension

We conducted an ablation study varying the SDAE latent dimension from 32 to 512 while keeping all other hyperparameters fixed (50 training epochs for both autoencoder and classifier).

Table 4: Effect of latent dimension on SDAE classification performance.

Latent Dim	Accuracy	F1 (Macro)
32	0.9295	0.8866
64	0.9268	0.8837
128	0.9286	0.8893
256	0.9272	0.8768
512	0.9272	0.8916

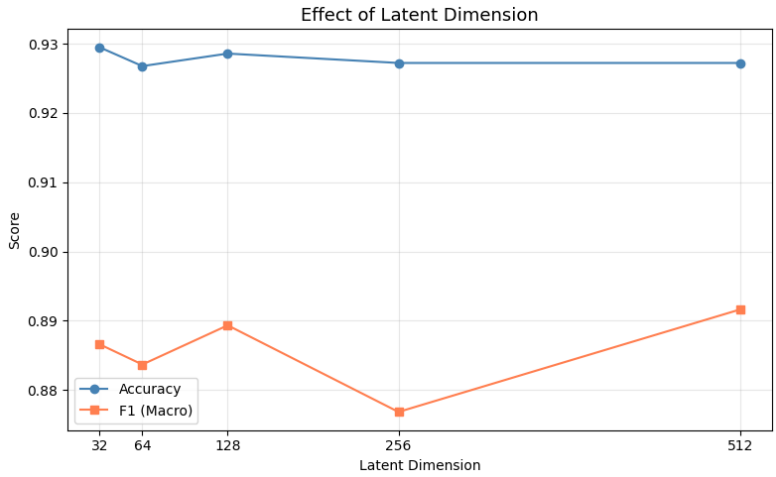

Figure 9: Effect of latent dimension on classification accuracy and F1-score.

As shown in Table 4 and Figure 9, classification accuracy remains remarkably stable across all latent dimensions (92.68–92.95%), suggesting that even a 32-dimensional bottleneck captures sufficient information for discriminating among 33 cancer types. The F1-score shows slightly more variability, with 128 and 512 dimensions performing marginally better. This stability indicates that the essential cancer-discriminating information in gene expression data is relatively low-dimensional, consistent with the biological understanding that cancer types are defined by the activity of a limited number of key pathways and gene programs.

5 Discussion

5.1 PCA Outperforming Autoencoders

A notable finding of this study is that PCA, a simple linear method, outperformed both non-linear autoencoder approaches for classification. This result, while perhaps surprising, is consistent with several observations. First, the gene expression data has already undergone log-transformation and batch correction, which may linearize many of the relationships between genes. Second, PCA produces orthogonal, decorrelated features that provide an ideal input for the downstream classifier, whereas autoencoder features may contain redundant or correlated information. Third, Chen et al. [1] achieved their 98% accuracy using autoencoders for binary cancer detection (tumor vs. normal), a simpler task than our 33-class classification.

It is important to note that classification accuracy is not the only metric of interest. The VAE's probabilistic latent space offers additional capabilities such as generating synthetic gene expression profiles and interpolating between cancer types, which PCA cannot provide. The SDAE's denoising property may make it more robust to noisy or low-quality samples in clinical settings.

5.2 Biologically Meaningful Classification Errors

The classification errors observed are biologically informative. The consistent confusion between colon and rectum adenocarcinoma reflects their shared molecular biology, and several studies have questioned whether they should be treated as distinct entities at the molecular level. Similarly, the overlap among kidney cancer subtypes (chromophobe, clear cell, and papillary) aligns with known molecular gradients within renal cell carcinomas. These results suggest that gene expression-based classifiers capture genuine biological similarity rather than arbitrary classification boundaries.

5.3 Comparison with Literature

Our results are broadly consistent with prior work. Chen et al. [1] reported 98% accuracy for cancer detection and over 90% accuracy for 18 of 32 cancer types using deep autoencoders. Mohammed et al. [4] achieved high accuracy using stacking ensemble approaches on TCGA data. Our PCA baseline accuracy of 94.67% across 33 cancer types compares favorably with these results, particularly given the larger number of classes. The classification patterns we observe—high accuracy for molecularly distinct cancers and lower accuracy for related subtypes—are consistent across studies.

5.4 Limitations and Future Work

Several limitations should be noted. First, we used a fixed architecture for both autoencoders; systematic hyperparameter optimization might improve their performance relative to PCA. Second, the feature selection step (top 5,000 genes by variance) is a simple filter that may not be optimal; more sophisticated gene selection methods could improve all models. Third, we did not explore fine-tuning the autoencoder and classifier end-to-end, which could improve the learned representations for the specific classification task.

Future work could explore attention-based architectures that learn to weight genes based on their importance for classification, graph neural networks that incorporate gene-gene interaction networks, integration of multi-omics data (DNA methylation, copy number variation, protein expression), and gene attribution analysis to identify which genes most influence classification decisions. These extensions could not only improve classification accuracy but also provide biological insights into the molecular determinants of cancer identity.

6 Conclusion

In this study, we systematically compared three dimensionality reduction methods—SDAE, VAE, and PCA—for classifying 33 cancer types from TCGA Pan-Cancer gene expression data. All three methods achieved classification accuracy above 92%, with PCA performing

best at 94.67%. The results demonstrate that gene expression profiles contain rich, discriminative information for cancer type identification, and that even linear dimensionality reduction captures the essential molecular signatures distinguishing different cancers. The classification errors provide biological insights into molecular similarity between related cancer types, and the ablation study confirms that the discriminative information is captured effectively even in low-dimensional latent spaces. These findings contribute to the growing body of evidence supporting machine learning approaches for molecular cancer classification and suggest directions for more sophisticated models that may further improve accuracy and interpretability.

References

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